Suppose you measure the side length of a cube and use it to calculate the volume of the cube. Suppose you measure the side length to be a with absolute uncertainty dx (we will use the words "uncertainty" and "error" interchangeably; we take $dx > 0$ here). Then the side length of the cube lies in the interval $(a-dx, a+dx)$. Since the measured value a of the side length is uncertain, the calculated volume will also be uncertain. Using your measured value and uncertainty, the maximum uncertainty in the volume is

$$|\Delta y| = |V(a+dx) - V(a)|,$$

so the volume of the cube lies in the interval

$$(V(a) - |\Delta y|, V(a) + |\Delta y|).$$

As we have seen in the previously, when dx is small, $\Delta y \approx dy = f'(a)dx$, so the interval above is approximately

$$(V(a) - dy, V(a) + dy).$$

We are also frequently interested in the *relative error* $\left|\frac{\Delta y}{y}\right|$ (that is, the size of the error relative to the size of the quantity being measured). The relative error is often quoted as the *percent error*: $\left|\frac{\Delta y}{y}\right| \times 100\%$. The relative error (or percent error) can be used as a measure of precision, with a smaller percent error meaning a more precise estimate. (E.g., is an error of 1 mile good? If you are measuring the length of the table it is horrible, but if you are measuring the radius of the Earth it is amazing.)

Example. Suppose the side length of a cube is measured to be 5 cm with an accuracy of 0.1 cm. Use differentials to estimate the interval in which the volume lies, as well as the percent error in the computed volume of the cube.

Solution. The volume is given by $V(x) = x^3$. The maximum error in the volume is approximately $dV = V'(x)dx = 3x^2dx = 3(5)^2(0.1) = 7.5$ cm^3. The volume therefore lies (approximately) in the interval $(125 - 7.5, 125 + 7.5) = (117.5, 132.5)$. The percent error is approximately

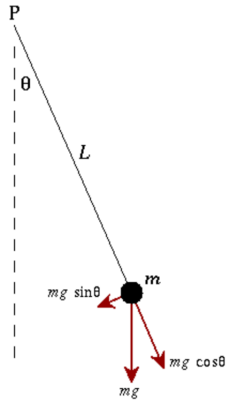
$$\begin{aligned}\frac{dV}{V} \times 100\% &= \frac{3x^2dx}{x^3} \times 100\% \\ &= \frac{3(0.1)}{5} \times 100\% \\ &= 6\%.\end{aligned}$$

1. An astronaut using a camera measures the radius of Earth as 4000 mi with an error of ± 80 mi. Use differentials to estimate the percentage error of using this radius measurement to calculate the volume of Earth, assuming Earth is a perfect sphere.

2. One side of a right triangle is known to be 20 cm long and the opposite angle is measured to be 30°, with a possible error of $\pm 1^\circ$. Use differentials to estimate the percent error in computing the length of the hypotenuse.

3. If an electrical current I passes through a resistor with resistance R, *Ohm's Law* states that the voltage drop is $V = IR$. If V is constant and R is measured with a certain error, use differentials to show that the relative error in calculating I is approximately the same (in magnitude) as the relative error in R.

4. (Optional) Consider the following free body diagram for a simple pendulum of mass m and length L. The gravitational force has been decomposed into components along the string and perpendicular to the string.



Newton's Second Law can be written as $\frac{d^2\theta}{dt^2} = -\omega^2 \sin(\theta)$, where $\omega = \sqrt{\frac{g}{L}}$, with $g \approx 9.8$ m/s^2 the acceleration due to gravity.

(a) The equation above cannot be solved in terms of elementary functions. Suppose we consider only small oscillations. Use the linear approximation of $\sin(\theta)$ at $\theta = 0$ to show that Newton's Second Law can be written for small oscillations as $\frac{d^2\theta}{dt^2} = -\omega^2\theta$.

(b) Show that the function $\theta(t) = A\cos(\omega t) + B\sin(\omega t)$, where the coefficients A and B are real constants, satisfies the equation in part (a).

(c) Solve for the constant coefficients A and B if $\theta(0) = \theta_0$ and $\frac{d\theta}{dt}(0) = 0$ and write the corresponding solution $\theta(t)$.

(d) The *period*, T, of the pendulum is the amount of time it takes to complete one full oscillation. Solve for T. How does T change if we double L? How does T change if we double θ_0?

(e) Use Desmos to find the range of θ values such that $\sin(\theta)$ differs from θ by less than 2%. State the range in degrees.